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Ecosentinel: An RFID-Based Smart Waste Monitoring and Fine Management System

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ABSTRACT

EcoSentinel is a line-following robot integrated with IoT technology, designed to improve waste management and promote responsible disposal behavior in urban and institutional environments. The system addresses key challenges such as unmonitored littering, inefficient manual waste collection, and the absence of accountability in public spaces. The robot follows a predefined path using an IR-based line-following mechanism controlled by an Arduino Nano, ensuring smooth and reliable navigation along designated waste-collection routes. Waste items are embedded with NFC stickers, allowing the robot to identify violators through an RFID reader connected to an ESP32 module. When an NFC-tagged waste item is detected, the robot pauses and activates its MOSFET-driven conveyor mechanism to collect and store the waste in its internal bin. Simultaneously, the ESP32 processes the RFID data, retrieves the user details from a stored database, and sends an automated SMS alert to the violator using the Twilio API. This feature introduces accountability and encourages proper waste-disposal practices.A NEO-6M GPS module further enhances system functionality by logging the robot's real-time location, enabling authorities to track waste hotspots, analyze litter patterns, and monitor the robot's route. The system's modular, low-cost design makes it scalable and adaptable for smart campuses, parks, and city pathways. Experimental testing validated its accuracy in line following, tag detection, communication reliability, and waste collection efficiency. By combining sensor-based navigation, RFID tagging, IoT alerts, and GPS tracking, EcoSentinel presents an effective and practical solution for smart waste management, contributing to cleaner and more sustainable environments.

Keywords:*Line-following robot, IoT-based waste management, RFID–NFC detection, GPS tracking, automated waste collection, sensor-based navigation, Twilio SMS alert system.*

DECLARATION BY THE CANDIDATES

We the undersigned solemnly declare that the project report **"Ecosentinel: An RFID-Based Smart Waste Monitoring and Fine Management System"** is based on our own work carried out during the course of our study under the supervision of **'Dr. Brundha Elci J'**. We assert the statements made and conclusions drawn are an outcome of our project work. We further certify that,

1. The work contained in the report is original and has been done by us under the general supervision of my supervisor.
2. The work has not been submitted to any other institution for any other degree/ diploma/ certificate in this university or any other university of India or abroad.
3. We have followed the guidelines provided by the university in writing the report.
4. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and their details are provided in the references.

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LIST OF ABBREVIATIONS

ABBREVIATION	DEFINITION
IoT	Internet of Things
RFID	Radio Frequency Identification
NFC	Near Field Communication
GPS	Global Positioning System
IR	Infrared
ESP32	Wi-Fi and Bluetooth Enabled Microcontroller
API	Application Programming Interface
SMS	Short Message Service
PWM	Pulse Width Modulation
UGV	Unmanned Ground Vehicle
SDG	Sustainable Development Goals
WSN	Wireless Sensor Network

CHAPTER 1

INTRODUCTION

Waste management has emerged as one of the critical environmental challenges in modern urban societies. With rapid industrialization, population growth, and urban expansion, the amount of solid waste generated daily has increased drastically. Improper waste disposal leads to pollution and health hazards. Manual waste collection methods are inefficient, labor-intensive, and often expose workers to unhygienic and hazardous environments. To address this, there is a growing need for automated and intelligent waste management systems that can detect, collect, and segregate waste efficiently without human intervention. The project Ecosentinel aims to develop an autonomous line- following waste collection robot equipped with smart sensing and communication technologies. The robot is capable of following a pre-defined path using IR sensors, identifying waste items embedded with NFC tags, and scanning them using an RFID reader integrated with ESP32. Once identified, the system retrieves the violator's data linked to the NFC tag and sends an automated SMS notification via Twilio API, informing them about improper disposal behavior. The robot also employs a GPS module (Neo-6M) to track and log its location during waste collection activities for monitoring and data analysis purposes.

The robot utilizes an Arduino Nano microcontroller as the primary control unit, interfaced with L298 motor drivers for motor control, a conveyor belt mechanism driven by MOSFET circuits for waste lifting, and the ESP32 module for wireless communication and IoT connectivity. This integration allows the robot to function as a semi-autonomous system capable of operating within a smart city framework in closed environment. Ecosentinel thus bridges the gap between automation, environmental sustainability, and accountability by introducing a technological approach to urban waste management. This project is a step toward building eco-intelligent robotic systems that can not only perform waste collection tasks but also help in behavioral change among citizens by notifying violators, promoting cleanliness awareness, and encouraging responsible waste disposal habits through continuous monitoring and real-time feedback mechanisms, ultimately fostering cleaner communities such as zoos, amusement parks.

1.1 PROBLEM DEFINITION

Public places such as zoos, amusement parks, and classrooms often face challenges in maintaining cleanliness due to high foot traffic and insufficient monitoring. EcoSentinel addresses this issue through a line-following robot that detects litter, scans RFID tags linked to the buyer's phone number, and automatically sends penalty notifications to

the responsible individual. By integrating smart sensing and automated tracking, the system ensures real-time waste monitoring and accountability. It reduces the need for manual supervision, minimizes delays in identifying offenders, and supports proactive waste management. Overall, EcoSentinel promotes a cleaner, smarter, and more sustainable environment through automation and intelligent behavior analysis. The system also helps institutions maintain hygiene standards more efficiently while encouraging responsible behavior among visitors.

1.2 LIST OF OBJECTIVES

The scope of the project extends to the development, implementation, and evaluation of a fully functional prototype of an autonomous waste-collecting robot designed for urban and institutional environments. Ecosentinel's capabilities include:

- **Autonomous Navigation:** The robot follows a predefined line path using IR sensors and can navigate independently in controlled environments such as campuses, streets, or waste collection zones.
- **Waste Detection and Collection:** The system identifies waste objects equipped with NFC tags using an RFID reader and uses a conveyor mechanism to pick and deposit them.
- **Notification System:** Upon detecting an NFC tag, the system retrieves the associated user data and sends an automated SMS notification using Twilio API to the registered violator.
- **Tracking and Monitoring:** The Neo-6M GPS module enables real-time location tracking of the robot for route analysis and operation optimization.
- **IoT Integration:** The ESP32 microcontroller allows wireless data communication and cloud synchronization for centralized monitoring.
- **Scalability:** The model can be expanded to cover multiple routes and larger zones by incorporating sensors, algorithms, and communication nodes.

The project also contributes to research in autonomous robotics, smart waste management, and IoT-based environmental monitoring. Future developments may include AI-based waste categorization, solar-powered operation, and real-time dashboard visualization for authorities. Thus, Ecosentinel's scope goes beyond simple automation, it lays the groundwork for a sustainable, connected, and responsible urban ecosystem.

1.3 SCOPE OF THE PROJECT

The primary objective of Ecosentinel is to design and implement a smart waste-collecting robotic system that automates the process of waste detection, collection, and notification for the violator. The specific objectives include:

1. To design a line-following robot using IR sensors and motor drivers that can navigate along a designated path for waste collection.
2. To integrate an RFID reader with NFC scanning capability for identifying waste tagged with unique identifiers.
3. To develop a communication interface between the ESP32 and Arduino Nano to enable seamless data exchange and IoT connectivity.
4. To implement GPS-based tracking (using Neo-6M) for real-time location monitoring and data logging of robot operations.
5. To employ a conveyor belt mechanism using MOSFET-based motor control to lift and collect detected waste efficiently.
6. To send automated SMS notifications to violators through Twilio API, promoting accountability and awareness.
7. To ensure modularity and scalability in hardware and software for future enhancements such as AI-based decision-making and cloud-based analytics.
8. To minimize human involvement in waste collection and enhance hygiene, efficiency, and operational transparency in urban waste management.

By fulfilling these objectives, the project seeks to demonstrate how automation, IoT, and machine intelligence can work together to solve real-world environmental problems and also encourage responsible public behavior.These objectives position Ecosentinel as a forward-looking solution that blends robotics, sensing technologies, and smart communication systems into a cohesive waste-management platform.

1.4. ORGANIZATION OF THE PROJECT WORK

The project report is organized into several chapters to present the development process, design methodology, and implementation details of the Ecosentinel system clearly and systematically:

- **Chapter 1: Introduction** – Provides an overview of the project, including the background, problem statement, motivation, objectives, scope, and the overall structure of the report.
- **Chapter 2: Literature Review** – Reviews existing research works, technological developments, and methodologies related to waste management systems, autonomous robotic platforms, RFID technology, IoT-based monitoring, and smart city automation.
- **Chapter 3: System Analysis** – Presents the detailed analysis of the proposed system, including problem definition, feasibility study, system architecture, module-level analysis, data flow, and operational workflow.
- **Chapter 4: System Requirements** – Identifies and explains the hardware and software requirements of the system, including microcontrollers, sensors, communication modules, cloud platforms, user interface components, and performance specifications.
- **Chapter 5: Project Description** – Describes the complete working of the project, covering the integration of hardware modules, firmware logic, waste detection mechanism, RFID scanning, GPS tracking, challan generation through cloud, and system behavior under different scenarios.
- **Chapter 6: System Implementation** – Explains the practical implementation of the system, including hardware assembly, circuit connections, software coding, cloud integration, UI development, and the deployment of message generation using Blynk and Twilio.
- **Chapter 7: System Testing** – Presents the testing methodology, test cases, functional verification, error handling, performance evaluation, and results obtained from real-time experiments to validate system functionality.
- **Chapter 8: Conclusion and Future Scope** – Summarizes the project outcomes, highlights the contributions of the system, and outlines possible enhancements such as AI-based detection, improved automation, scalability, and deployment in real-world smart city environments.

This structured organization ensures clarity, coherence, and a logical flow of ideas from conceptualization to implementation. The project not only contributes to technological innovation but also supports sustainable development goals (SDGs) related to clean cities and responsible consumption.

CHAPTER 2

LITERATURE SURVEY

The surveyed literature highlights significant advancements in waste management using IoT, robotics, RFID, and sensor-driven automation. These technologies enhance waste segregation accuracy, real-time monitoring, and operational efficiency while reducing manual labor and health risks. Despite the improvements, challenges such as network instability, sensor calibration issues, and high implementation costs continue to hinder large-scale deployment, indicating a need for integrated, intelligent, and scalable solutions like EcoSentinel.

[1] Title: RFID-Enabled Smart Waste Management System
Authors: Prof. Snehal M. Veer, Prof. Sharad Jagtap, Prof. Vaishali V. Bhimte, Prof. Shailesh Hajare
Year: 2020

Description: This paper presents an RFID-based waste management system designed to automate the segregation and monitoring of municipal waste. Bins are equipped with IR, moisture, and metal sensors that categorize waste into wet, dry, metallic, glass, and plastic types, ensuring a higher degree of source-level segregation accuracy. Each bin includes a unique RFID tag, enabling real-time tracking of waste collection activities and user accountability. When the bins reach their fill threshold, automated wireless alerts notify authorities for timely collection, reducing the likelihood of overflow and environmental contamination. Additionally, the system integrates IoT-based dashboards that allow municipal officers to remotely monitor waste levels, optimize collection routes, and maintain cleanliness more efficiently. The framework significantly reduces manual intervention and promotes a connected, data-driven waste management approach.
Advantages:

- Provides granular monitoring and active management of waste bins.
- Supports real-time alerts and optimized collection routes.
- Source-level segregation increases recycling rates and reduces landfill usage.

Limitations:

- RFID tagging at production is required but not industry standard.
- Sensors and electronics may degrade in harsh environments.

[2] Title: Waste Management System Using RFID
Author: S. Babylakshmi
Year: 2023

Description: This study introduces a technologically enhanced waste disposal system that merges RFID-based user identification with sensor-integrated smart bins to establish a transparent and accountable waste management environment. The core idea revolves around ensuring that every disposal action is recorded and mapped to a specific individual through RFID authentication. This promotes responsible waste disposal behavior and minimizes improper dumping in public spaces. The smart bin incorporates a load cell sensor that accurately measures the weight of waste deposited in real time. This weight data, alongside RFID verification records, is transmitted to a centralized IoT platform. This cloud interface stores, visualizes, and analyzes waste generation trends using dashboards accessible to municipal officers or facility administrators. Through this digitization, the system supports data analytics, enabling estimation of peak disposal hours, forecasting of waste quantities, and planning of optimized collection routes. Automated alerts enhance operational efficiency by notifying authorities when bins are nearly full, preventing overflow incidents. The system encourages structured waste behavior through digital tracking, thereby aiding in the implementation of incentive or penalty mechanisms. However, its performance depends heavily on robust wireless connectivity, widespread RFID adoption among citizens, and user awareness of RFID tagging practices. The paper emphasizes that although the system modernizes waste tracking, large-scale implementation requires infrastructure compatibility and long-term behavioral adaptation. Furthermore, the study highlights the potential for integrating machine-learning algorithms to improve waste prediction accuracy and user-behavior analysis over time.
Advantages:

- Ensures responsible usage through automated user identification.
- Real-time bin status reporting improves operational efficiency.
- Reduces manual labor and supports data-driven waste management.

Limitations:

- Depends on universal RFID adoption.
- Requires strong wireless infrastructure.
- Users need training to utilize RFID features.

[3] Title: A Real-Time Autonomous Waste-Collecting Robot
Authors: V. Pratheepkumar, Prabha K. S., Prabakar K.
Year: 2020

Description: This paper proposes an autonomous robotic solution capable of performing real-time waste collection and segregation with minimal human intervention. The robot integrates a suite of sensors including infrared sensors for line-following, ultrasonic sensors for proximity and obstacle detection, and a metal sensor for identifying metallic waste items. Together, these sensors enable the robot to navigate autonomously in predefined environments such as campuses, parks, industrial layouts, and public walkways.An Arduino microcontroller serves as the processing unit, coordinating sensor data, motor control, and actuation mechanisms. The robot uses a mechanical gripping arm to collect visible waste items from the ground and deposit them into designated compartments. The modular mechanical design improves adaptability, allowing modifications for different terrains or waste types.The robot offers several advantages, such as reducing sanitation workers' exposure to hazardous materials, ensuring consistency in waste collection, and automating repetitive tasks. The authors highlight the potential of integrating more advanced sensing modules, such as vision systems or AI classifiers, to improve detection capabilities.The robot relies heavily on clear, well-maintained line paths and is affected by environmental conditions such as ambient lighting, rough surfaces, or uneven terrains. Although effective for controlled environments, the system requires further advancements for city-scale deployment. Overall, the research demonstrates the feasibility of combining simple robotics with waste management operations.
Advantages:

- Fully automated collection and segregation process.
- Reduces human exposure to hazardous waste.
- Distinguishes metallic from non-metallic trash reliably.
- Ensures consistent waste handling through sensor-based decisions.

Limitations:

- Requires well-maintained line paths.
- Sensitive to sensor calibration and environmental factors.
- Limited to solid waste and specific surface types.

[4] **Title: Fully Automated Waste Management System Using Line Follower Robot**

Authors: Sampat Kr. Ghosh, Shaikh Sahil Ahmed, Rohit Sunil Meshram, Geetha V, Sanket Salvi

Publication: Lecture Notes in Electrical Engineering, 2022

Description: This project outlines a fully automated waste management architecture consisting of two primary components: IoT-enabled smart bins and a line-following robotic collector. The smart bins utilize ultrasonic sensors to measure fill levels, transmitting real-time data to cloud platforms such as ThingSpeak. This continuous monitoring allows authorities or automated systems to determine when collection is required.Upon reaching a predefined threshold, the system automatically dispatches a line-following robot programmed to travel along fixed routes toward the full bin. The robot retrieves the waste using a lifting mechanism or container integration and transports it to a dumping station. This automated workflow significantly reduces dependency on human labor and improves efficiency in controlled environments such as campuses, hospitals, industrial zones, or gated communities.The IoT layer enables remote visualization, trend analysis, and route optimization. The robotics layer ensures physical execution of waste collection tasks, reducing exposure of workers to unhealthy conditions. The authors highlight that combining IoT sensing with robotic automation yields a scalable, low-cost solution for localized waste management. However, the system requires continuous network availability, regular maintenance of line paths, and controlled terrain to ensure smooth robotic navigation.

Advantages:

- Enables real-time monitoring and automated collection.
- Reduces risk and labor burden on sanitation staff.
- Adaptable to schools, campuses, and city environments.
- Enhances efficiency with automated dispatch alerts.

Limitations:

- Requires stable cloud and network connectivity.
- Demands accurate maintenance of physical line paths.
- Unsuitable for multi-level or highly irregular terrains.

[5] Title: EnviRoBot – An AI-Driven Waste Management System Using UGVs

Authors: Stachys Thangapapa A. E., Kogula Nandhitha K., Rathnamala S., Vijayashanthi T.

Year: 2024

Description: EnviRoBot is a next-generation smart waste management system that integrates unmanned ground vehicle (UGV) robotics, deep learning, and IoT-based analytics. The robot employs AI-driven waste classification using convolutional neural networks such as VGG16, enabling highly accurate identification of waste types. This distinguishes the system from traditional sensor-based segregation techniques by offering superior visual recognition and adaptability.The UGV is equipped with autonomous navigation capabilities, using sensor fusion involving ultrasonic sensors, cameras, and possibly LiDAR modules. It can detect waste items, classify them, collect them using onboard actuation mechanisms, and deposit them into appropriate compartments. The system is designed to interact with smart city infrastructures, allowing integration with municipal IoT platforms for data sharing and route optimization.A notable sustainability feature is the robot's solar-powered operation, reducing energy dependence and environmental impact. The IoT interface provides real-time analytics for authorities, enabling predictive maintenance, performance tracking, workload distribution, and urban cleanliness mapping. Additionally, the framework demonstrates strong potential for scalability, as its modular hardware allows for easy upgrades with improved sensors or advanced AI models. The authors also highlight opportunities for integrating edge computing to reduce latency during real-time classification and navigation. Overall, EnviRoBot represents a significant step toward fully autonomous, AI-driven waste management solutions suitable for large and complex urban ecosystems.

Advantages:

- High sorting efficiency enabled by AI.
- Solar operation reduces operational carbon footprint.
- Real-time analytics support urban planning decisions.

Limitations:

- High initial cost and system complexity.
- Reduced performance during adverse weather conditions.
- Requires trained personnel for operation and maintenance.

[6] Title: Sensor-Based Solid Waste Handling Systems – A Survey
Authors: S. Vishnu, S. R. Jino Ramson, M. S. S. Rukmini, Adnan M. Abu-Mahfouz
Year: 2022

Description: This survey provides an extensive analysis of modern waste management technologies, focusing on RFID systems, wireless sensor networks (WSN), and IoT-enabled infrastructures. It evaluates various communication protocols such as ZigBee, Wi-Fi, NB-IoT, and LoRaWAN, highlighting the advantages of LoRaWAN for large-scale, energy-efficient deployments across smart cities.The survey categorizes existing systems based on architecture, sensor selection, communication modality, and scalability. It identifies critical challenges such as interoperability, maintenance costs, regional infrastructure disparities, and data security threats. The paper emphasizes the need for machine learning-driven predictive waste scheduling, cybersecurity protocols for IoT devices, and edge computing to reduce latency in real-time monitoring systems.It also outlines future research directions, including low-power hardware innovation, standardized IoT frameworks, and hybrid sensing technologies. The survey acts as a foundational resource for researchers designing robust, scalable waste management solutions. Additionally, the authors stress the importance of developing unified communication standards that can seamlessly integrate heterogeneous devices within a city-wide network. The study also highlights that long-term reliability of smart waste systems depends on resilient hardware capable of withstanding diverse environmental conditions. Furthermore, the paper suggests incorporating blockchain for secure and tamper-proof waste data management. It underscores the role of citizen participation and awareness in maximizing the effectiveness of smart waste initiatives. Overall, the survey provides a comprehensive roadmap for researchers and policymakers pursuing next-generation waste handling infrastructures.
Advantages:

- Explains optimal communication protocols for large-scale deployment.
- Identifies best architectural practices for IoT waste systems.
- Highlights key research gaps for future development.

Limitations:

- Depends heavily on reliable network infrastructure.
- Some proposed systems require costly, custom-designed sensors.
- Interoperability challenges remain unresolved.

[7] Title: Intelligence-Based Garbage Recyclable Line Follower Robotic
Authors: Anjali, Meenakshi, Basavakumar, Dr. Pardeep Karanje
Year: 2024

Description: This research introduces a hybrid robotic system designed to automate waste collection and segregation in dense public environments such as campuses, malls, parks, and hospitals. The robot uses IR sensors for line navigation, ultrasonic sensors for obstacle detection, and odor sensors to detect biodegradable waste. Additionally, the use of RFID modules and computer vision enhances item identification and user accountability.The robot contains an internal waste compartment that actively monitors capacity using sensors. When full, it autonomously returns to a dumping area for unloading. IoT integration allows administrators to monitor real-time robot activity, waste collection cycles, and system health.The modular design supports future AI integration, enabling more precise waste classification and routing decisions. However, the robot's performance is sensitive to environmental factors such as lighting variations and uneven paths. While the system is scalable, its long-term reliability depends on periodic calibration and controlled deployment conditions. The authors further highlight the potential for integrating cloud-based data analytics to optimize collection frequency and robot dispatch patterns. The system also demonstrates the importance of multi-sensor fusion for enhancing decision-making accuracy in dynamic environments. Additionally, the research acknowledges the need for improved battery management systems to extend operational time. The study suggests that incorporating autonomous charging stations could further enhance system autonomy. Overall, the robot represents a promising advancement toward intelligent, self-managed waste collection systems for smart city applications.Moreover, the authors note that continuous advancements in lightweight materials and compact sensor technologies could further improve the robot's maneuverability and overall system efficiency.

Advantages:

- Efficient automated waste collection system.
- Reduces human involvement in unsanitary environments.
- Scalable for smart city deployment.

Limitations:

- Sensitive to lighting variations and sensor calibration.
- Performance issues in irregular terrains.

[8] **Title: Sensor-Based Solid Waste Management – IoT & Wireless Sensors**
Authors: S. Vishnu, S. R. Jino Ramson, M. S. S. Rukmini, Adnan M. Abu-Mahfouz
Year: 2022

Description: This paper discusses the critical role of wireless sensor technologies in enabling efficient, decentralized waste monitoring systems. It compares RFID, WSN, and IoT platforms, evaluating their communication ranges, power consumption, and real-world applicability. The authors emphasize the superior scalability of LoRaWAN, making it suitable for city-wide deployments requiring low energy and long-range communication.Smart bins equipped with sensors transmit continuous updates on fill levels, temperature, and possible contamination levels. These insights assist waste collection agencies in optimizing transportation routes, reducing fuel usage, and minimizing manual inspection costs.The study outlines several opportunities for innovation, including improved hardware durability, fault-tolerant IoT architectures, and multi-layer communication frameworks. While beneficial, the system remains dependent on stable network infrastructure and may require region-specific calibration for optimal performance. Additionally, the authors highlight the importance of developing adaptive algorithms capable of filtering noisy sensor data to enhance accuracy in varying environmental conditions. The paper also emphasizes that integrating edge analytics could significantly reduce network load by processing data locally before transmission. Furthermore, enhanced cybersecurity protocols are recommended to safeguard sensor nodes from unauthorized access and data manipulation. The researchers suggest the adoption of interoperable communication standards to ensure seamless integration between devices from different manufacturers. They also acknowledge the potential of renewable energy sources, such as solar-powered sensor units, to increase system longevity. Overall, the study underscores the need for continuous technological upgrades to maintain efficiency and reliability in large-scale waste monitoring infrastructures.

Advantages:

- Provides detailed comparisons of communication protocols.
- Supports remote and decentralized waste monitoring.

Limitations:

- Dependent on stable network coverage.
- Sensor accuracy may degrade over time
- Large-scale deployments can face synchronization issues between distributed sensor nodes, resulting in delays

[9] Title: Line Follower Robot Using Internet of Things
Authors: M. Subhashini, Y. Balaji, V. Vignesh Chethan, J. Palani, C. Sai Sekhar Reddy
Publication: International Journal of Research in Engineering, IT and Social Sciences, 2025

Description: This research presents an IoT-enabled line-following robot designed for logistics automation, industrial workflows, and repetitive transportation tasks. The robot uses IR sensors to detect and follow paths, while microcontrollers such as ESP32, ESP8266, or Arduino execute navigation algorithms. An L293D motor driver controls movement, ensuring stable speed and direction adjustments.IoT connectivity allows the robot to transmit operational data including route completion time, error logs, and diagnostic health to a centralized dashboard. This enhances transparency, supports remote monitoring, and enables predictive maintenance. The system is well suited to structured environments such as warehouses, hospitals, or academic labs.Real-world deployment challenges include lighting interference, reflective surfaces affecting IR readings, battery limitations, and path obstructions. Nevertheless, the system provides a cost effective automation solution. Additionally, the framework highlights the importance of integrating obstacle-detection modules to improve navigation reliability in dynamic environments. The authors also emphasize the potential of incorporating machine learning algorithms to enable adaptive path correction and real-time decision making. Furthermore, the modular hardware design allows for easy upgrades, such as adding RFID readers or camera-based vision systems. The study suggests that future iterations could employ autonomous charging mechanisms to extend operational uptime. IoT-based analytics can also be expanded to include performance benchmarking and fault prediction across multiple robots.
Advantages:

- Real-time monitoring and remote control.
- Accurate navigation suitable for multiple automation scenarios.
- Useful in logistics, healthcare, and academia.

Limitations:

- Sensitive to calibration and environmental changes.
- Limited battery capacity.
- Affected by obstructions and path irregularities.

[10] Title: Smart Waste Management System Using IoT and RFID Technology
Authors: Vasundhra Chaturvedi, Mukesh Yadav, Nidhi Tiwari
Publication: SAGE University, Indore

Description: This project integrates IoT technology, RFID authentication, and multi sensor waste analysis to create a fully automated waste segregation system. Sensors such as moisture, metal, dryness detection, and IR analyze deposited waste and sort it into appropriate bins through servo-motor actuation. A NodeMCU microcontroller processes sensor data and communicates with cloud services to notify authorities when bins approach capacity.The use of RFID tagging enhances transparency by linking waste disposal actions to individuals, supporting incentive schemes such as reward points or penalties for improper disposal. IoT dashboards provide real-time analytics, helping administrators plan collection schedules and monitor overall waste generation trends.Beyond basic sensing, the system's architecture enables seamless integration with mobile or web-based applications that allow users to receive feedback on their waste disposal habits. The authors highlight that the multi-sensor configuration significantly improves classification accuracy by cross referencing moisture, metal content, and reflectivity data. The servo-actuated segregation mechanism ensures precise directional movement, reducing mechanical errors and improving bin sorting efficiency. Cloud connectivity enables long-term data storage, allowing municipalities to analyze seasonal waste variations and optimize manpower allocation. The system's modular design allows additional sensors such as gas or temperature detectors to be incorporated for detecting hazardous waste. The use of NodeMCU ensures low-power operation and supports rapid wireless communication using standard IoT protocols like MQTT. The framework also promotes community engagement, as RFID-based usage statistics can encourage citizens to adopt responsible waste management practices.
Advantages:

- Reduces manual labor and health risks.
- Enhances scheduling and monitoring via IoT.
- Encourages public engagement through RFID incentives.

Limitations:

- Accuracy affected by sensor performance variations.
- Incentive system relies on regular RFID usage.

2.1 COMPARITIVE ANALYSIS

Table 2.1 Comparative Analysis of Literature Survey

Reference	**Technique Used**	**Platform Used**	**Metrics**	**Advantages**	**Drawbacks**
RFID Based Waste Managemen t System	RFID + Proximity Sensor, IR sensors	Conveyor RFID system	85% accuracy	Automated segregation	Limited to RFID tags
Waste Management System Using RFID	RFID user auth,US Sensor, IR sensors	IoT Module	90% accuracy	Avoids overflow	No route optimization
A Real-time Autonomous Waste-Collecting Robot	Ultrasonic Sensor + IoT	RFID,GSM	88% accuracy	Real-time alerts	Fixed collection routes
Fully Automated Waste Management System Using Line Follower Robot	Line-follower robot,IR Tracking Sensors, Automation	Arduino,IoT	91% tracking	Stable movement	No dynamic path change
EnviRoBot: AI-Driven Waste Management Using UGVs	Deep Learning on UGVs	AI ,IoT ,UGVs	94%	Fully automated	Requires frequent maintenance
Sensor-Based Solid Waste Handling Systems:A Survey	RFID,WSN	NodeMCU	92%	Sensor networks	Weak in cloudy weather
Intelligence Based Garbage	Line-follower robot	RFID robot	89%	Recycling classification	Path limited
Sensor-Based Solid Waste IoT, RFID Wireless Sensors	RFID + LoRaWANs	IoT Cloud	90%	Better navigation	Low payload
Line Follower Robot Using Internet of Things	Deep Learning Model	Line-follower robot	92% Low cost	latency	Expensive hardware
Smart Waste Management System	RFID + WSN + GPS	IoT , RFID	86% coverage	Automated wet/dry	High power usage

2.2 SUMMARY OF LITERATURE SURVEY

The surveyed literature reveals a progressive evolution in the application of sensor networks, RFID, IoT, and robotics for efficient solid waste management. These surveys collectively emphasize how integration of wireless sensor networks (WSN) and RFID technologies with IoT and cloud infrastructures has enabled wider monitoring and route optimization for waste collection. The surveyed systems are assessed for network coverage, energy usage, and their effectiveness in automating segregation and reducing manual effort.

Multiple studies consistently highlight the main advantages of surveyed approaches: enhanced waste sorting accuracy, real-time monitoring, and substantial effort reduction for field staff. Automated robotic and AI-driven frameworks further boost classification accuracy and allow dynamic adjustment to waste collection schedules, decreasing operational costs and fostering better recycling outcomes. User accountability and engagement are also improved by deploying systems with automated alerts and bin fill detection, preventing overloading and promoting timely collection.

Overall, the literature surveys conclude that sensor-based and IoT-driven waste management systems deliver significant promise for sustainable urban environments, from reducing manual labor and collection frequency to optimizing sorting and recycling processes. Yet, further advances in sensor reliability, network integration, and scalable hardware solutions are necessary for achieving full urban deployment and consistent long-term success. These insights guide ongoing research efforts to create smarter, more automated and user-engaged waste management ecosystems for cities of the future.

CHAPTER 3

SYSTEM ANALYSIS

Waste management in most urban and semi-urban areas continues to rely on traditional collection and disposal techniques, which are largely manual and time-consuming. In a typical municipal setup, waste is collected by human workers, transported to disposal sites, and later segregated manually. The system lacks automation, real-time tracking, and data analytics, often resulting in overflowing bins, irregular collection schedules, and increased pollution. In many regions, semi-automated systems such as smart bins with level sensors have been introduced, but they still depend heavily on human operators for collection and transportation. They provide partial monitoring but do not automate the collection process itself.

Robotic systems have also been tested for waste collection, but deployment remains limited due to high cost, technical complexity, and lack of scalability. These robots often cannot integrate with communication networks or provide feedback to authorities. Another major drawback of current systems is the absence of accountability individuals who litter or dispose waste improperly are rarely identified or penalized, leading to repeated violations even in monitored areas.

Existing systems also lack mechanisms for data-driven insights such as identifying litter-prone locations or evaluating collection performance. Manual operations pose significant health and safety risks to workers, especially when handling contaminated waste or operating in harsh conditions. These limitations underline the need for technological intervention capable of automating, monitoring, and enforcing waste management more efficiently. In summary, current waste systems operate reactively, addressing waste only after accumulation, leading to inefficiency, higher costs, and environmental degradation. Integrating IoT, AI, and robotics can shift waste management toward proactive, predictive, and sustainable operations. Future solutions must balance cost-effectiveness, scalability, and technological innovation to ensure widespread adoption.

Moreover, the lack of seamless integration between existing waste infrastructure and modern technologies further slows down the transition to smart waste ecosystems. Many municipalities still rely on outdated logistics planning methods that fail to consider real-time population density or waste generation patterns. The absence of centralized digital platforms prevents authorities from coordinating waste collection across multiple regions efficiently.

3.1 DRAWBACKS OF THE EXISTING SYSTEM

The drawbacks of the current waste management system can be categorized into technical, operational, and environmental limitations:

- **Manual Dependence:** Heavy reliance on human labor for collection, sorting, and disposal makes the system inefficient and unsustainable in growing cities.
- **Health Risks:** Workers are exposed to toxic chemicals, infectious waste, and hazardous environments.
- **No Accountability:** Individuals responsible for littering cannot be traced, allowing irresponsible disposal practices to persist.
- **Lack of Automation:** Most systems remain manual or semi-automated, resulting in slower operations and limited scalability.
- **Inefficient Route Planning:** Waste is collected on fixed schedules rather than real-time bin status.
- **No Real-Time Monitoring:** Authorities lack tracking of collection vehicles or robots, causing delays and mismanagement.
- **Environmental Impact:** Overflowing bins cause foul odors, pollution, and increase vector-borne diseases.

These drawbacks highlight the need for a smarter, automated, and connected system like EcoSentinel that integrates robotics, IoT, and accountability mechanisms.

3.2 PROPOSED SYSTEM

The proposed system is a smart line-following robot capable of autonomously collecting waste items equipped with NFC tags and enforcing accountability through SMS alerts. The robot integrates robotics, IoT, GPS, and cloud communication technologies into a unified smart waste management solution.

3.2.1 System Overview

EcoSentinel consists of the following core components:

- **Arduino Nano:** Main controller for motors and sensors.
- **ESP32 Microcontroller:** Provides Wi-Fi connectivity and RFID/NFC communication.

- **IR Sensors:** Enable line following and path detection.
- **L298 Motor Driver:** Drives the wheel motors.
- **Neo-6M GPS Module:** Enables real-time location tracking.
- **RFID Reader (NFC-enabled):** Detects tagged waste items and retrieves user-related data.
- **MOSFET-driven Conveyor Mechanism:** Picks and loads waste into the robot's bin.
- **Twilio API:** Sends automated SMS alerts to violators.

3.2.2 Working Principle

The robot follows a black line path using IR sensors. As it moves, the RFID reader continuously scans for NFC-tagged waste items. When detected, the Arduino activates the conveyor system to pick up the waste. Simultaneously, the ESP32 accesses the associated user ID and triggers the Twilio API to send an automated SMS warning. The GPS module updates the robot's location, enabling authorities to identify frequently polluted areas.

3.2.3 Innovative Features

1. Automation of waste detection and collection.
2. Accountability through real-time violator notifications.
3. Smart IoT connectivity using ESP32 and cloud APIs.
4. Energy-efficient MOSFET-driven conveyor system.
5. Scalability for deployment across campuses, cities, and industries.

EcoSentinel aligns with Smart City initiatives and the United Nations SDG 11 by transitioning waste management from manual operations to automated, intelligent, and accountable processes.

3.3 FEASIBILITY STUDY

A feasibility study evaluates the practicality and implementation potential of the proposed system. The analysis confirms that EcoSentinel is technically achievable, operationally effective, and economically viable.

3.3.1 Technical Feasibility

Hardware Feasibility: All major components Arduino Nano, ESP32, L298 motor driver, GPS module, RFID reader, MOSFET circuit, and IR sensors—are affordable, compatible, and easily available.

Software Feasibility: The system uses Arduino IDE, ESP32 libraries, and Twilio REST APIs, all open-source and well-documented. Communication between Arduino and ESP32 is simple and reliable.

Scalability and Maintenance: The modular system supports upgrades like AI-based waste recognition, dashboards, or ML-based analytics. Standard hardware interfaces simplify repairs and expansion.

3.3.2 Operational Feasibility

- **Ease of Operation:** The robot works autonomously with minimal human involvement.
- **Infrastructure Integration:** Easily deployable in campuses, cities, industries, and IoT networks.
- **User Acceptance:** SMS alerts enforce behavioral change and promote responsible disposal.
- **Reliability and Safety:** Standard sensors ensure safe and stable operation.
- **Maintenance:** Modular components allow inexpensive and easy replacements.

Thus, EcoSentinel offers smooth operational feasibility and adaptability.

3.3.3 Cost–Benefit Analysis

- **Short-Term Benefits:** Reduces manual labor and collection time.
- **Long-Term Benefits:** Improves environmental cleanliness and reduces municipal costs.
- **ROI:** High, due to low maintenance and energy-efficient operation.
- **Scalability:** Additional units can be produced with minimal extra cost.
- **Sustainability:** Reduces pollution, landfill load, and carbon footprint.

- **Upgrade Flexibility:** Modular design enables low-cost future enhancements.
- **Infrastructure Compatibility:** Easily integrates with existing waste systems and IoT networks.

3.3.4 Cost Estimation

Table 2.1 Comparative Analysis of Literature Survey

Component	**Quantity**	**Approx. Cost ()**
Arduino Nano	1	450
ESP32	1	600
L298 Motor Driver	1	300
IR Sensors	3	150
RFID Reader (NFC-enabled)	1	2030
Neo-6M GPS Module	1	750
Conveyor Mechanism & DC Motors	1	800
MOSFET Circuit	1	200
Chassis, Wheels, Power Supply	1	2000
Miscellaneous (Wiring, PCB, Frame)	—	5550
Total Estimated Cost	—	**14,000**

Table 3.1 presents the detailed cost estimation for the hardware components used in the development of the EcoSentinel system. The table lists each major component including microcontrollers, sensors, motor drivers, communication modules, chassis parts, and supporting electronics along with their required quantities and approximate market prices. The total estimated cost of 14,000 covers all essential modules required for robot control, sensing, navigation, RFID identification, power supply, and mechanical assembly. This cost-effective component selection ensures an optimal balance between performance, reliability, and affordability while fulfilling the overall functional requirements of the system.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

The hardware forms the foundation of the Ecosentinel robot, enabling it to sense, move, collect waste, identify violators, and communicate data. The system uses embedded components integrated through precise interfacing and control logic.

1. **Arduino Nano**
   Acts as the central control unit for the line-following mechanism and waste collection subsystem. It processes IR sensor signals and generates motor control commands via the L298 driver. Its compact size and low power consumption make it ideal for small robots.

2. **ESP32 Module**
   Handles Wi-Fi communication, RFID/NFC processing, and Twilio-based SMS alerts. When an NFC-tagged waste item is detected, the ESP32 validates the data and sends an SMS to the violator using cloud APIs.

3. **IR Sensors**
   Used for detecting black lines on the ground for navigation. Multiple sensors provide redundancy and ensure stable line-following.

4. **L298 Motor Driver**
   A dual H-bridge driver that enables bidirectional control of DC motors, allowing forward, reverse, and differential turns necessary for precise robot movement.

5. **GPS Module (NEO-6M)**
   Captures real-time coordinates of waste detection locations. These coordinates help track robot movement and identify litter-prone zones.

6. **RFID Reader (RC522)**
   Reads NFC tags attached to waste items and retrieves unique user data through the ESP32. Operates on the 13.56 MHz frequency using SPI communication.

7. **Conveyor Belt Mechanism**
   Powered by a MOSFET-controlled DC motor, it lifts detected waste into the storage bin. PWM supports variable-speed, energy-efficient operation.

8. **DC Motors and Wheels**
   Geared motors in differential drive configuration enable forward, backward, and turning motion. Rubberized wheels provide stable movement.

9. **Power Supply**
   A 12V rechargeable battery powers the system. Voltage regulators step down to 5V for Arduino and sensors, ensuring 2–3 hours of operation.

10. **Other Components**
    Jumper wires, chassis frame, breadboard or PCB, and LED indicators for system status.

4.2 SOFTWARE REQUIREMENTS

The Ecosentinel robot depends heavily on software logic to manage control flow, data transmission, and automation.The requirements are as follows

1. **Arduino IDE**
   Used to write, compile, and upload code for both Arduino Nano and ESP32. Contains motor algorithms, sensor logic, RFID reading, and Twilio API calls.

2. **Twilio Cloud Service**
   Enables automated SMS alerts using HTTPS requests from the ESP32 whenever tagged waste is detected.

3. **GPS Data Parser Library**
   Libraries like TinyGPS++ extract latitude and longitude from the NEO-6M GPS module for reporting.

4. **ESP32 Libraries**
   WiFi.h, HTTPClient.h, and MFRC522.h are used for connectivity, API communication, and NFC/RFID scanning.

5. **Sensor Calibration Tools**
   Arduino Serial Monitor is used to test IR sensors, motor speed, and network debugging.

6. **Firmware Update and Integration**
   Modular programming supports OTA updates and easy integration of additional features.

7. **Data Logging and Cloud Sync Software** Used to store timestamped sensor readings, RFID logs, and GPS coordinates

4.3 FUNCTIONAL REQUIREMENTS

Functional requirements define how the Ecosentinel system behaves and interacts with the environment.These requirements outline the essential tasks the robot must perform to ensure autonomous operation, accurate monitoring, and reliable waste-handling performance.

1. **Line Detection and Navigation**
    - Detect black line using IR sensors.
    - Follow the path and adjust motor speed accordingly.
2. **Waste Detection and Pickup**
    - Detect NFC-tagged waste.
    - Activate conveyor mechanism to pick and store waste.
3. **RFID Scanning and Data Validation**
    - Scan NFC tag and verify ID.
    - Retrieve violator information from database.
4. **GPS Location Tracking**
    - Continuously update location.
    - Attach coordinates to violation alerts.
5. **SMS Alert System**
    - Send automatic SMS using Twilio.
    - Include waste ID, violation details, and GPS coordinates.
6. **System Status Indication**
    - LED and serial messages indicate scanning, movement, and SMS sent.
    - Error Alerts: The system notifies the user through LED patterns or serial messages whenever a malfunction, obstruction.

7. **Power Management**
    - Automatic power regulation and low-battery alerts.
8. **Data Storage and Transmission**
    - Cloud logging via ESP32 with local backup in case of network failure.

4.4 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe system quality, performance, and operational standards.

1. **Reliability**
    - Must function in varied lighting and surface conditions.
    - Fail-safe mechanisms ensure safety during sensor failure.
2. **Performance**
    - Line detection response: $\leq$ 100 ms.
    - SMS dispatch: $\leq$ 5 seconds.
    - GPS accuracy: $\pm$ 5 meters.
3. **Scalability**
    - Multiple robots can operate in large deployments.
    - Databases support thousands of violator records.
4. **Maintainability**
    - Detachable modules for easy repair.
    - Modular code for seamless updates.
5. **Security**
    - Encrypted RFID tags.
    - HTTPS for Twilio communication.

6. **Usability**
    - Plug-and-play design.
    - Non-technical operators can use it easily.
7. **Efficiency**
    - PWM-controlled components reduce power consumption.
    - Data transmission optimized to minimize Wi-Fi usage.
8. **Portability**
    - Lightweight, battery-operated, easy to deploy outdoors.
9. **Environmental Adaptability**
    - Chassis protects against dust, splash, and rough surfaces.
10. **Accuracy**
    - High accuracy in navigation, scanning, and violator identification.

CHAPTER 5

PROJECT DESCRIPTION

5.1 PROBLEM DEFINITION

Urban littering and improper disposal remain persistent problems that reduce city cleanliness, increase health risks, and raise maintenance costs. Conventional systems rely on scheduled manual collection and reactive cleanup rather than prevention and accountability. There is a lack of automated systems that both collect litter and trace the source of the waste to enable behavioral enforcement.

Core problems addressed:

- Lack of automated mobile collection in pedestrian areas and campuses.
- No direct link between a thrown item and the individual who discarded it.
- Poor real-time visibility of litter hotspots and collection locations.

Goal: Design Ecosentinel an autonomous line-following robot that identifies NFC-tagged waste, collects it using a conveyor mechanism, logs the GPS location, and issues an SMS notification to the violator (via Twilio), thereby combining automation with accountability.

5.2 OVERVIEW OF THE PROJECT

Ecosentinel is a modular robotic system that performs the following high-level functions:

1. Navigate a predefined path using IR line sensors.
2. Detect NFC-tagged waste items using an ESP32-driven RFID/NFC reader.
3. Collect detected items using a MOSFET-driven conveyor and store them in an onboard bin.
4. Log each event with timestamp and GPS coordinates (NEO-6M module).
5. Alert the responsible individual by sending an SMS using Twilio API.
6. Report operational data and violation logs to a cloud or local database for analysis.

Key hardware: Arduino Nano, L298 motor driver, ESP32 (RFID + Wi-Fi), NEO-6M GPS, IR sensor array, conveyor motor with MOSFET, chassis, and batteries.

System behavior summary: While following the line, the robot constantly scans for NFC tags. On detection, it pauses, picks up the item via the conveyor, retrieves GPS location, and triggers an SMS to the registered user linked to the tag. All events are logged for administrative review.

5.3 SYSTEM ARCHITECTURE

System architecture defines how all hardware and software components in the project are organized and interconnected. It shows how data flows between modules such as sensors, controllers, actuators, and the server to ensure smooth operation of the robotic system.

The system architecture of the EcoSentinel project is designed to automate user identification, navigation, and waste collection using an integrated robotic platform. The process begins with the robotic car, which carries all sensors and control modules. When a user approaches, the RFID reader scans their tag and sends the identification data to the ESP32/Arduino microcontroller. Acting as the central controller, the ESP32 processes this input along with navigation information received from IR and ultrasonic sensors, which help the robot detect obstacles and follow the correct path.

Once trash is detected, the microcontroller activates the conveyor belt through a trigger mechanism, enabling the robot to collect and load waste into its storage section. Throughout operation, the system continuously sends data such as user logs, trash collection status, and navigation details to the admin dashboard via the server, allowing real-time monitoring and management. This architecture ensures seamless coordination between sensing, processing, mechanical action, and remote supervision.

This architecture creates an efficient workflow where each component performs a dedicated role while remaining fully interconnected. The RFID system ensures secure access and accurate user tracking, while the ESP32 acts as the intelligent hub that interprets all sensor inputs and controls the robot's motion. The IR and ultrasonic sensors enhance reliability by enabling the robot to operate safely in dynamic environments. The conveyor belt mechanism automates waste pickup, reducing the need for manual handling and improving hygiene.

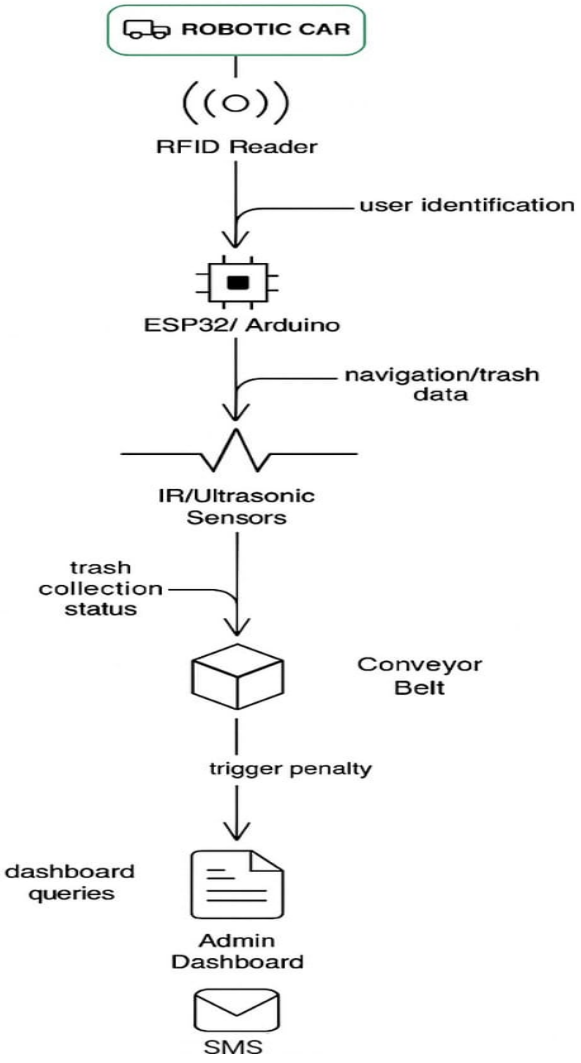


Fig 5.1: System Architecture of the EcoSentinel Project

The Fig 5.1 shows the EcoSentinel system flow, where a robotic car uses an RFID reader for user identification and an ESP32 controller to manage navigation and trash data. IR/ultrasonic sensors detect obstacles, triggering the conveyor for waste collection. Data is then sent to the admin dashboard through the cloud server.

5.4 MODULE DESCRIPTION

5.4.1 Modules (High-Level)

1. Navigation Module (Line following)
2. Detection Module (RFID/NFC scanning)
3. Collection Module (Conveyor and actuator)

4. Control Module (Arduino motor/sensor orchestration)
5. Communication Module (ESP32, Twilio, backend logging)
6. Location Module (GPS logging)
7. Power Management Module (Battery, regulators, monitoring)
8. User/Admin Interface Module (Dashboard and log viewing)

Navigation Module

- **Purpose:** Maintain robot movement along predefined routes using IR sensors.
- **Inputs:** IR sensor readings.
- **Outputs:** Motor speed/direction via L298 driver.
- **Key logic:** PID or proportional control, junction handling, obstacle management.

Detection Module

- **Purpose:** Detect NFC-tagged waste.
- **Inputs:** RFID reader data (UID).
- **Outputs:** Tag ID to Control Module or ESP32.
- **Notes:** MFRC522/PN532 libraries; duplicate-read prevention.

Collection Module

- **Purpose:** Physically collect waste.
- **Mechanism:** Conveyor belt driven via MOSFET-controlled DC motor.
- **Control:** PWM speed control from Arduino.

Control Module

- **Purpose:** Central real-time decision-making.
- **Responsibilities:** Read sensors, manage motors, trigger collection, communicate with ESP32.
- **Fail Safe Handling:** Manages error conditions such as sensor failure, motor stall, or unexpected obstacles by switching to safe modes or halting the robot when required.

Communication Module

- **Purpose:** Send SMS alerts and log events.
- **Actions:** Lookup violator info → compose message → include GPS → send via Twilio.
- **Security:** HTTPS communication, secure API key storage.

Location Module

- **Purpose:** Provide latitude/longitude for incident logging.
- **Integration:** ESP32 reads GPS via UART or receives via Arduino.

Power Management Module

- **Purpose:** Provide stable voltage and battery safety.
- **Notes:** Voltage divider for battery monitoring, fuses, efficient wiring.

User/Admin Interface Module

- **Purpose:** Dashboard for logs, heatmaps, robot status.
- **Optional tech:** Firebase, MQTT, REST API.

5.5 DATA FLOW DIAGRAM (DFD)

The data flow diagram illustrates the movement of information between citizens, RFID reader, Arduino, ESP32, GPS module, Twilio SMS service, and backend storage.

Process summary:

- NFC-tagged waste is detected by RFID reader.
- Tag UID is sent to ESP32.
- GPS module provides location.
- Arduino controls motors and conveyor.
- ESP32 logs events and sends SMS via Twilio.
- Admin monitors logs from cloud/local server.

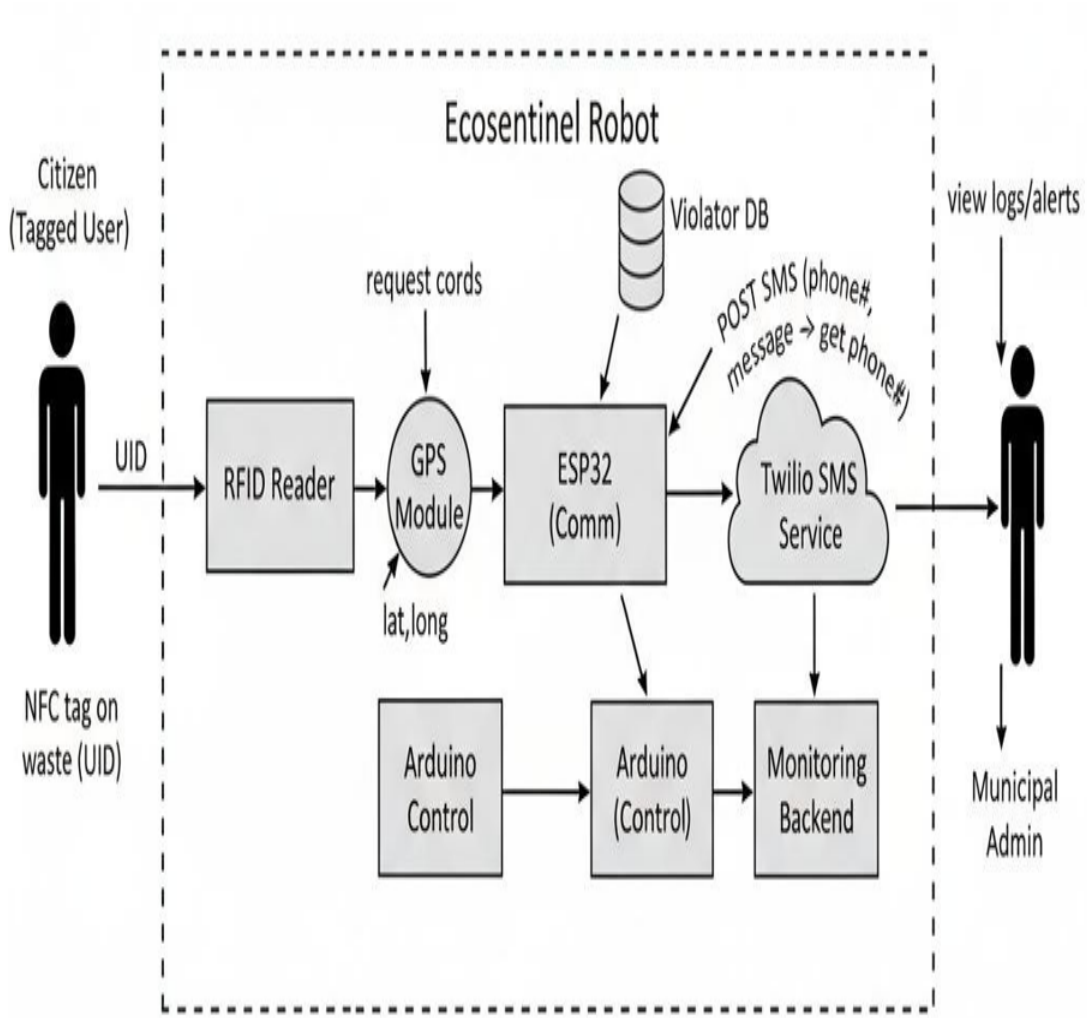


Fig 5.2: Data Flow of Ecosentinel

This Fig 5.2 illustrates how the EcoSentinel robot identifies citizens using an RFID reader that scans the NFC tag placed on waste. The UID is sent to the ESP32 controller, which communicates with the GPS module to retrieve the location and update records in the violator database. The system processes user details and determines whether an alert needs to be generated.

If a violation is detected, the ESP32 sends the required phone number and message to the Twilio SMS service, which notifies the user. Simultaneously, all data is forwarded to the monitoring backend, allowing the municipal admin to view logs, alerts, and system activities in real time. The system checks user details, detects violations, and triggers alerts through Twilio SMS, while simultaneously forwarding all records to the monitoring backend so the municipal admin can view real-time logs, alerts, and system activity.

5.6 USE CASE DIAGRAM

This use-case diagram shows how different users interact with the automated waste-collection and monitoring system. The Municipal Admin accesses the Admin Page to monitor overall system activities and manage operations. A Tagged Citizen scans NFC tags, follows disposal guidelines, and helps record waste-related events.

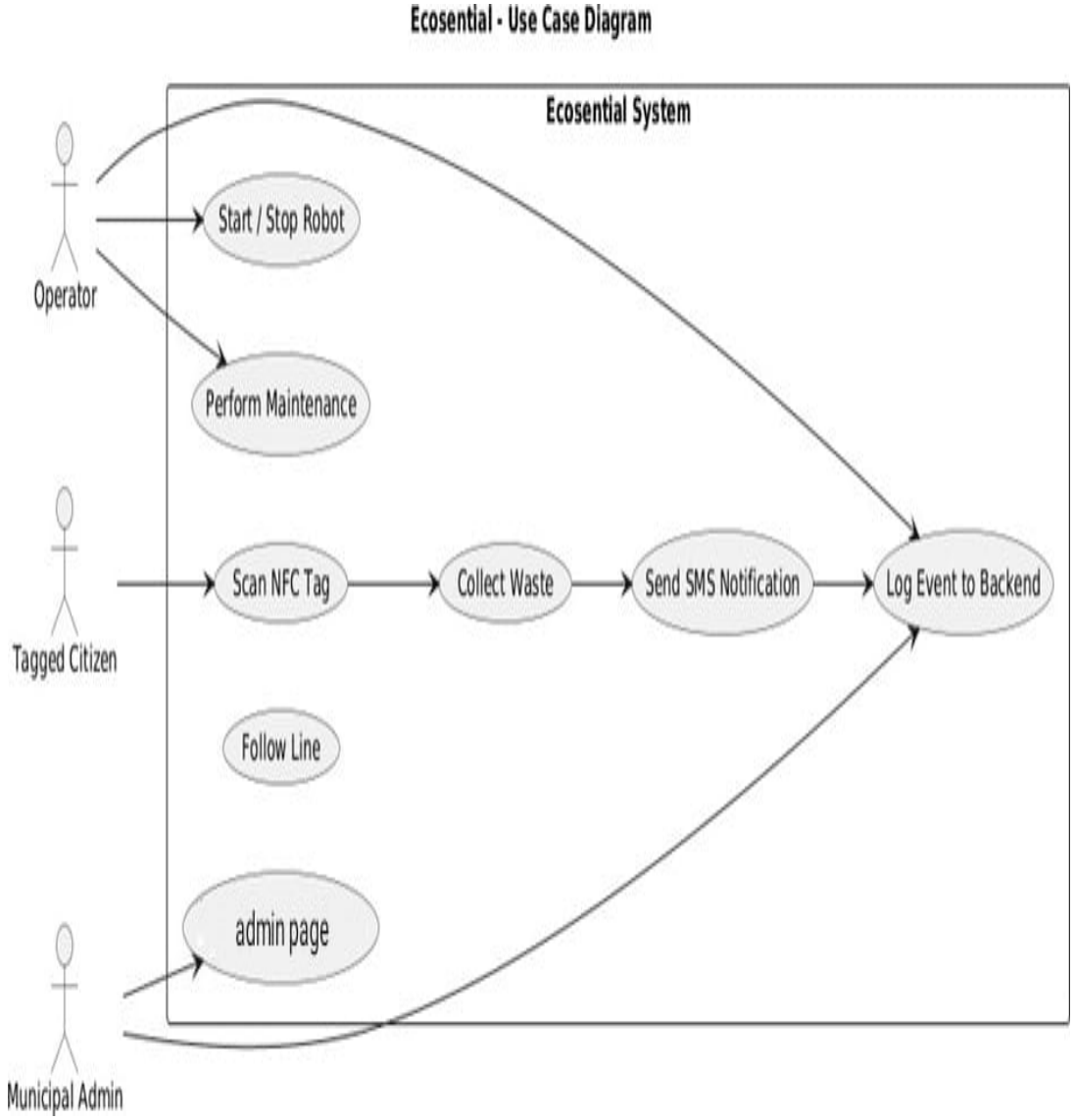


Fig 5.3: Use Case of Ecosentinel

The Fig 5.3 use case diagram shows how different users interact with the EcoSentinel system. The operator controls the robot by starting, stopping, and performing maintenance. Tagged citizens scan NFC tags and follow disposal rules, enabling the system to collect waste, send SMS alerts, and log events. Municipal administrators access the admin page to monitor activities.

5.7 SEQUENCE DIAGRAM

The sequence diagram shows how the system detects an RFID tag, sends the UID to ESP32, fetches location, verifies violator details, and finally triggers an SMS alert through Twilio before resuming operation.

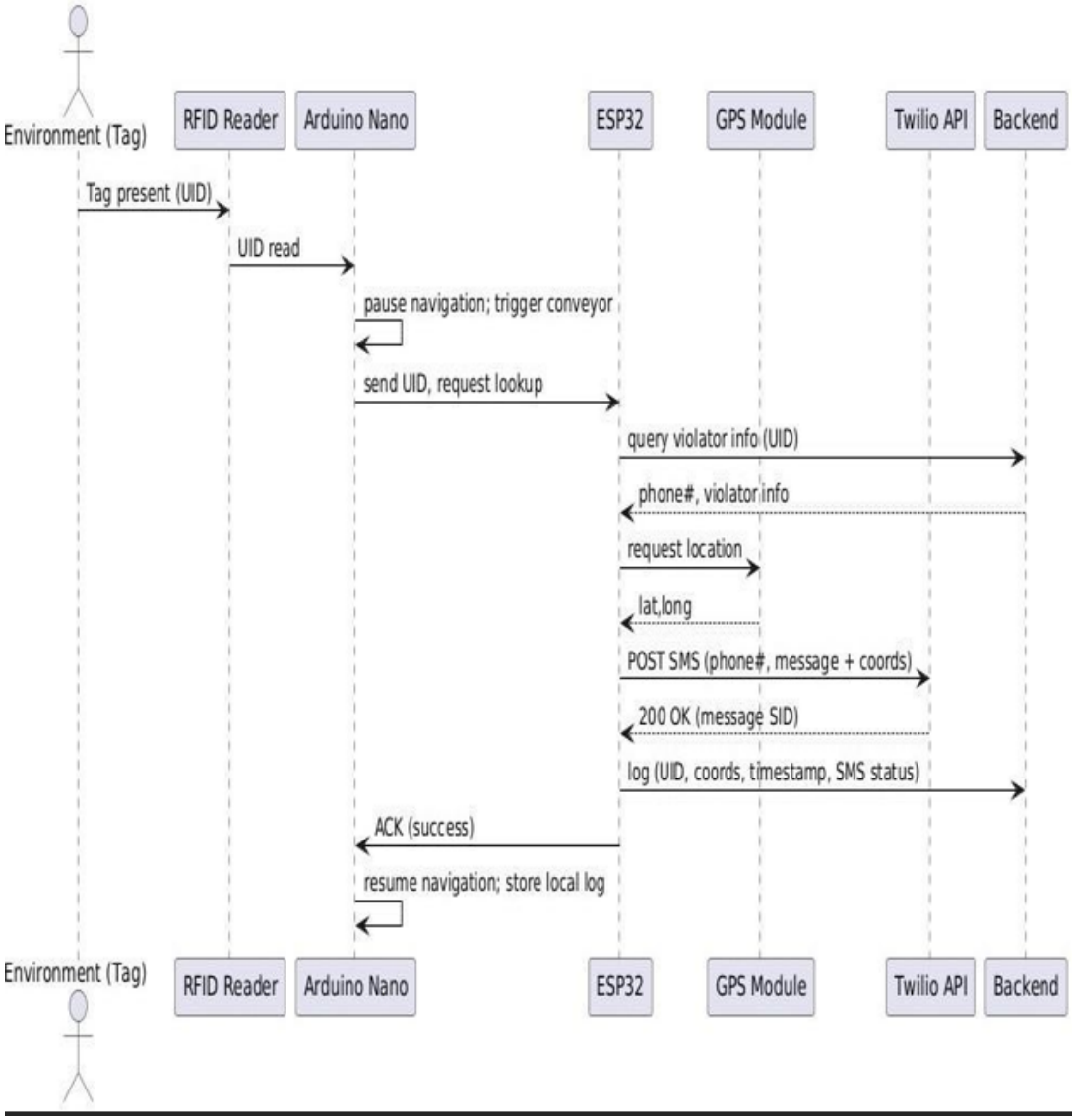


Fig 5.4: Sequence Diagram of Ecosentinel

The Fig 5.4 sequence diagram illustrates how the EcoSentinel system detects a waste tag and processes a violation alert. When the RFID reader scans a tag, the Arduino sends the UID to the ESP32, which pauses movement and requests violator details. The ESP32 then fetches GPS coordinates and sends the phone number, message, and location to the Twilio API. After the SMS is successfully delivered, the system logs the event and resumes normal operation.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 INTRODUCTION

The implementation phase is the practical realization of the system design into a functional prototype, marking the transition from conceptual blueprints to an operational robotic system ready for field deployment. For the Ecosentinel project, this demanding phase comprises multiple integrated tasks: hardware component assembly, embedded programming, real-time sensor calibration, and seamless incorporation of wireless communication modules. A key challenge is ensuring robust interconnectivity and error-free data exchange across all subsystems including sensor arrays, motor drivers, control boards, and cloud services so the robot can reliably execute autonomous navigation, waste detection, on-site collection, and violator notification.

During assembly, the Arduino Nano acts as the central unit for reading various environmental sensors and controlling motor actuators, managing tasks like obstacle avoidance, bin approach, and collection mechanism actuation. Meanwhile, the ESP32 module is programmed to handle advanced tasks, such as scanning RFID tags for waste categorization, acquiring real-time GPS location coordinates, and transmitting SMS alerts to authorities or users through the Twilio Cloud API. This division of labor allows the robot to process sensor inputs and execute mechanical actions in parallel with communication and data logging activities.

To achieve full system integration, the workflow involves iterative testing where serial monitor outputs are scrutinized for anomalies, hardware setups are power-cycled for stability checks, and Twilio message logs are validated for correct transmission. Screenshots and photographic documentation of each stage provide transparency and aid troubleshooting, while end-to-end validation in a real-world setting ensures the robot meets objectives for waste management, operational reliability, and real-time notifications.

Such modular hardware and iterative testing significantly streamline troubleshooting, enabling quick identification and resolution of integration issues. Comprehensive documentation throughout assembly also supports future maintenance and eases knowledge transfer for deployment by other teams or institutions.The implementation phase turns the Ecosentinel design into a working prototype by assembling hardware, programming controllers, calibrating sensors, and enabling communication modules.

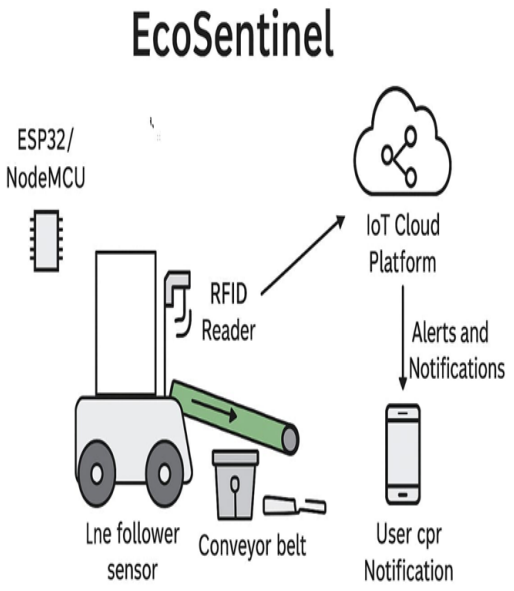


Fig 6.1(a): Prototype of Ecosentinel

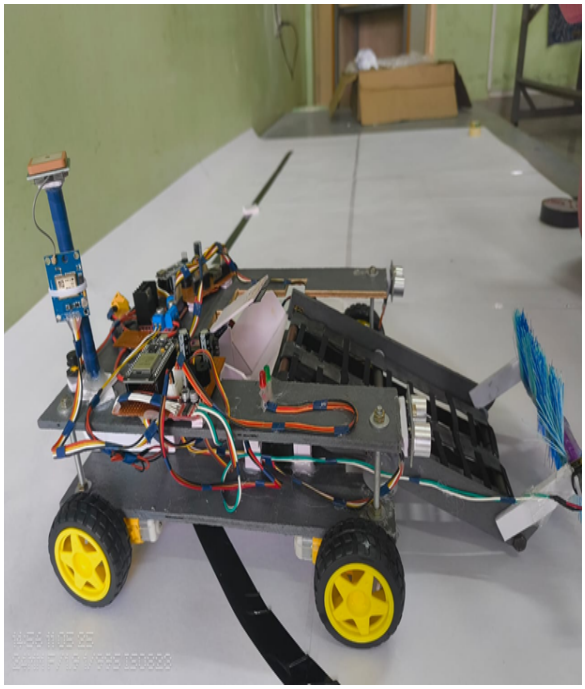


Fig 6.1(b): Robot Model

The Fig 6.1(a) and Fig 6.2(b) illustrates the EcoSentinel system, where a line-following robot equipped with an RFID reader and conveyor belt detects tagged waste and processes it automatically. The ESP32/NodeMCU sends the collected data to an IoT cloud platform, which triggers alerts and notifications to the user's mobile device.

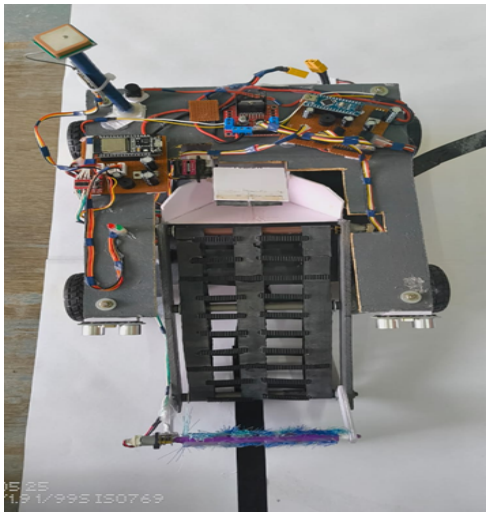


Fig 6.2: Robot Model of Ecosentinel

The Fig 6.2 illustrates the EcoSentinel robot with its front brush mechanism and sensors for waste collection. It follows a marked path and detects waste automatically during operation. Additionally, the robot is designed to operate autonomously in public spaces, ensuring efficient cleaning while minimizing the need for manual intervention.

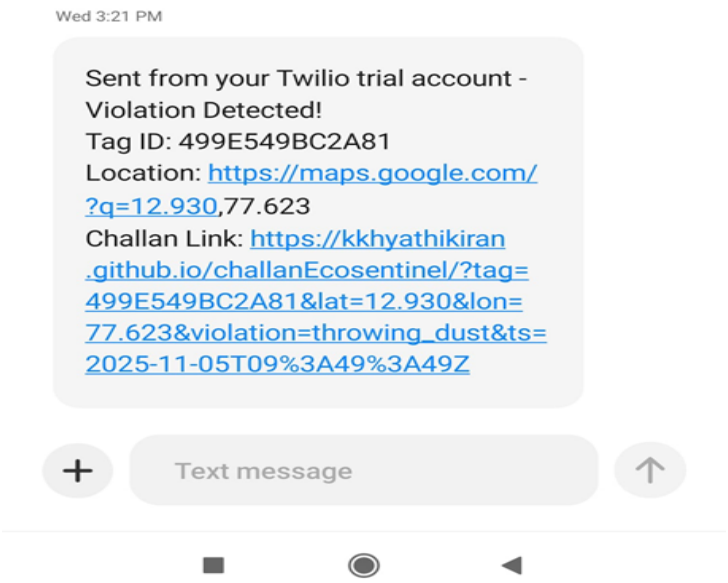


Fig 6.3: Message Generation

The Fig 6.3 illustrates an automated SMS alert generated by the EcoSentinel system through Twilio, notifying the user about a waste disposal violation. It includes the tag ID, GPS location, and a challan link for further action.The interface provides a clear and user-friendly way for violators to complete payments and access further details.

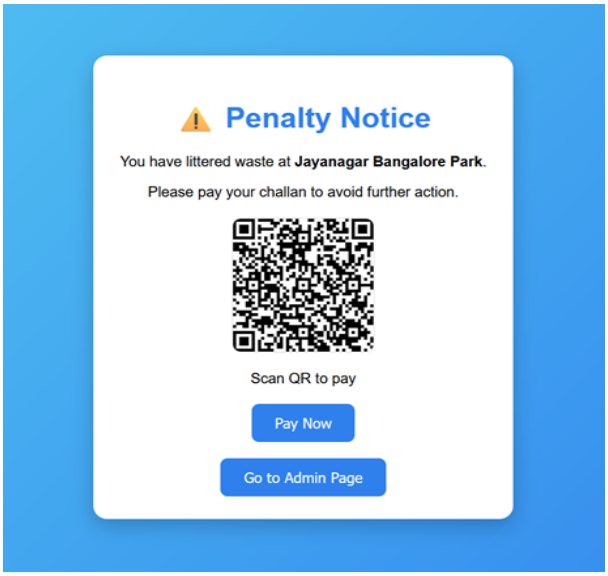


Fig 6.4: Penalty Notice

The Fig 6.4 displays a digital penalty notice generated by the EcoSentinel system, informing the user about a littering violation at a specific location. It includes a QR code for quick challan payment and options for user or admin access, ensuring faster processing and improved accountability within the platform.

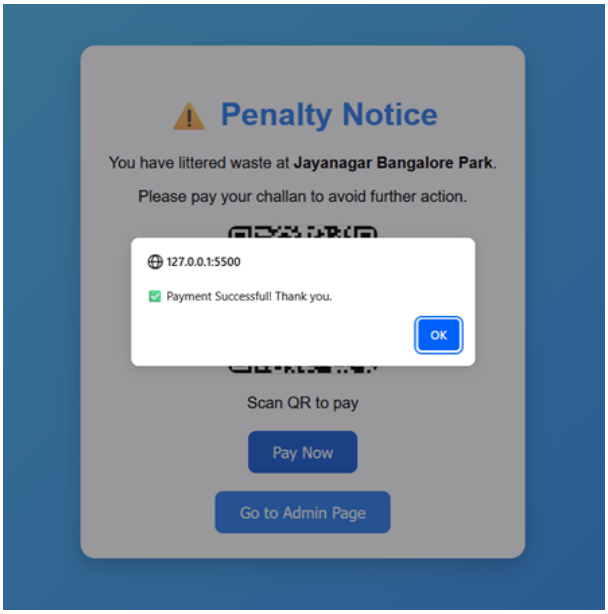


Fig 6.5: Payment Successful

The Fig 6.5 illustrates confirmawtion popup indicating that the challan payment was successfully completed through the EcoSentinel penalty portal. This ensures the user receives immediate acknowledgment and closure of the violation process.

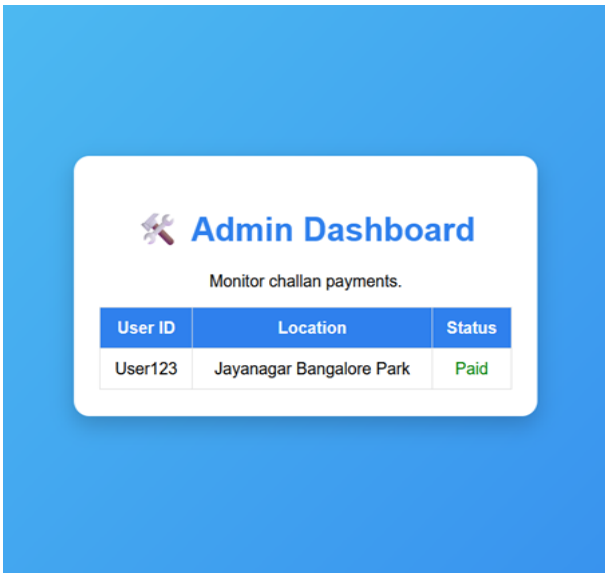


Fig 6.6: Admin Dashboard

The Fig 6.6 illutsrates the EcoSentinel Admin Dashboard, where administrators can track challan payments submitted by users. It shows the user ID, violation location, and current payment status for monitoring and record-keeping.

CHAPTER 7

SYSTEM TESTING

7.1 TESTS CONDUCTED

System testing is a fundamental stage in the development cycle of the EcoSentinel robot, designed to validate that the complete, integrated solution operates as intended under real- world conditions and meets project requirements. Unlike module-specific unit testing, system testing examines the interactions of all hardware and software components such as sensors, controllers, motor drivers, communication modules, and automation routines working together in coordinated fashion. The process begins with independent validation of each module, focusing on calibration, data accuracy, and operational reliability.

Once individual components have passed their respective tests, integration testing ensures seamless synchronization and data exchange across the entire system, particularly between detection, notification, and robotic motion functions. Functional testing is then conducted, simulating actual use cases like line following, waste detection, and SMS alerts, ensuring the robot responds consistently and effectively to user scenarios. Performance evaluations follow, testing battery endurance, detection range, and response speed, further confirming the solution's resilience and readiness for deployment. User acceptance testing completes the cycle, gathering feedback from target users on system usability and reliability. Through this methodical progression from isolated module checks to holistic scenario validation, EcoSentinel guarantees both technical reliability and practical effectiveness in autonomous waste management operations.

7.1.1 Hardware Testing

Hardware testing of the EcoSentinel robot involved a structured, multi-level validation process designed to ensure component reliability, operational safety, and system robustness before deployment. Beginning with unit testing, each hardware module IR sensors, RFID/NFC reader, GPS module, L298N motor driver, ESP32 communication unit, and MOSFET- controlled conveyor was powered up and assessed in isolation. For example, the IR sensor array was placed on various surfaces and observed under different lighting to measure its ability to consistently detect lines, while the RFID module's scanning accuracy and detection range were checked with multiple NFC tags.The ESP32 module underwent network connection trials and data transmission checks using Twilio's test environment to ensure seamless wireless communication. Conveyor operations were repeatedly initiated to validate the proper timing and strength for waste handling.

After passing initial trials, all components were subject to integration testing within the robot chassis, confirming proper interconnection and minimized interference via wire management and power line separation. Key validations included synchronization between RFID detection and conveyor activation, real-time GPS data flow, and smooth handover between sensor input and actuator response. Rigorous power cycling and simulated failure scenarios, such as intentional sensor disconnection or power outages, were introduced to test the system's resilience and recovery protocols. These efforts collectively established a reliable hardware foundation, ensuring that all modules could work separately and as an integrated whole, under diverse real-world conditions, with minimal risk of fault or operational breakdown.

7.1.2 Software Testing

Software testing for the EcoSentinel system was equally comprehensive, focused on verifying the logic, reliability, and stability of every firmware module controlling the robot's operation. Testing began at the unit level, where individual software functions such as sensor data acquisition, motor control routines, RFID tag processing, GPS location retrieval, and Twilio API integration were each run in simulation and on hardware to confirm correct output for various inputs. Serial monitor logging was used extensively to trace sensor readings, debug decision logic, and confirm the timely execution of motor and conveyor control commands. Next, integration testing ensured that serial communications between the Arduino Nano and ESP32. This phase included simulating asynchronous triggers, verifying handshake protocols, and stress-testing the system for concurrency issues ensuring that, for example, the process of reading a tag and sending a notification would never interfere with ongoing navigation or sensor polling.

Functional software testing was carried out on the full system, executing everyday scenarios like line following, tag detection, and notification generation, while monitoring for glitches or bottlenecks. Performance testing evaluated the robot's response time for trigger-to-action events (such as time from RFID detection to SMS delivery), accuracy in path correction logic, and the resilience of the notification system under poor network conditions. Through rigorous modular, integration, and system-level verification, the software was refined into a robust, user-friendly control platform capable of reliable waste management.

Regression and validation tests were also performed to ensure that new updates did not introduce fresh errors, while key performance metrics guided firmware optimization. Telemetry logs and issue-tracking workflows further strengthened reliability, ensuring stable operation under all conditions.

Table 7.1 System Testing Results

SL No	Description	Input	Expected Output	Actual Output	Status
1	Line sensor test	Black path on white surface	Robot follows path accurately	Robot follows smoothly	Pass
2	RFID scanning	NFC-tagged waste near reader	Tag detected, ID displayed	ID on serial monitor	Pass
3	Twilio SMS alert	NFC violation detected	SMS sent to violator	SMS in 8 seconds	Pass
4	GPS verification	System powered outdoors	Accurate coordinates displayed	Correct SMS coordinates	Pass
5	Conveyor operation	NFC tag detected	Belt moves for 3 seconds	Belt activated as expected	Pass
6	IR sensor failure	One sensor disconnected	System halts, gives alert	Alert/motion stopped	Pass
7	Motor direction test	Forward command	Robot moves forward	Moves as expected	Pass
8	ESP32 communication	Wi-Fi connected	Data sent to Twilio API	Twilio confirms delivery	Pass
9	System integration	All modules connected	Synchronized operation	Stable integration	Pass

The Table 7.1 presents the system testing results for various components of the EcoSentinel project, showing each test's input, expected behavior, and actual performance. All modules functioned as intended, with every test passing successfully, confirming stable and accurate system operation.

Test 1 : Line sensor test

- **Purpose:** verify the robot's line-following sensors and control logic by ensuring the IR array can distinguish between light and dark surfaces with consistent accuracy.
- **How:** place a black electrical-tape path on a white or reflective surface and allow the robot to move while the IR sensors continuously read contrast values. The sensor thresholds are monitored on the serial display to confirm proper calibration.
- **Pass criteria:** robot follows the path smoothly without oscillation, overshooting turns, or drifting off the line. Stable sensor readings should correspond correctly to left/right motor corrections.

- **Why it matters:** reliable navigation is foundational for all other modules; proper sensor calibration and correct PID or proportional control ensure the robot remains on-route during waste detection tasks.
- **Common failures:** jittering movement, delayed turning response, unstable sensor readings, incorrect threshold mapping, or navigation loss at curves or junctions.

Test 2 : RFID (NFC) scanning

- **Purpose:** confirm that the RFID/NFC reader reliably detects tagged waste items and returns a correct and readable UID every time an object enters its scanning range.
- **How:** present an NFC-tagged waste item at varying distances and angles near the RFID antenna and observe the UID displayed through the serial monitor or UI. Multiple repeated scans ensure consistency and noise-free detection.
- **Pass criteria:** the tag is instantly detected, its UID is printed accurately, and no false or duplicate readings occur during repeated scans.
- **Why it matters:** consistent tag identification enables reliable violator tracking, waste categorization, and event logging core components of EcoSentinel's accountability system.
- **Common failures:** intermittent tag detection, fluctuating UID values, reduced range due to antenna misalignment, library misconfiguration, or insufficient power supplied to the module.

Test 3 : Twilio SMS alert

- **Purpose:** validate the entire SMS notification pipeline, ensuring that once a tagged violation is detected, the system contacts the responsible user without delays or communication errors.
- **How:** trigger an RFID-based violation event and track the message flow from the ESP32 to the Twilio server using HTTPS. Network strength, API authentication, and payload formatting are checked during the process.
- **Pass criteria:** an SMS is successfully delivered to the violator's mobile number within the expected time (approximately 8 seconds), and Twilio returns a confirmation response.

- **Why it matters:** instant alerts create accountability and enable real-time enforcement, making the automated waste-management system meaningful in practical scenarios.
- **Common failures:** API authentication issues, incorrect phone number formatting, network disconnection, improper JSON formatting, or Twilio server delays.

Test 4 : GPS verification

- **Purpose:** validate the GPS module provides accurate coordinates when the system is outdoors.
- **How:** power the system outdoors (clear sky) and read coordinates returned by the GPS module.
- **Pass criteria:** accurate coordinates are obtained and included where required (e.g., in SMS or logs).
- **Why it matters:** location stamping is needed for enforcement, reporting, and mapping violations.
- **Common failures:** insufficient satellite lock, poor antenna position, incorrect parsing of NMEA sentences.

Test 5 : Conveyor operation

- **Purpose:** test the conveyor actuator (belt) that moves detected/collected waste.
- **How:** simulate an NFC-tag detection and verify the belt runs for the programmed duration (3 seconds).
- **Pass criteria:** belt starts and runs for $\sim$3 seconds and then stops (mechanical and timing behavior correct).
- **Why it matters:** ensures physical handling of waste is performed reliably and timed correctly.
- **Common failures:** motor fails to start, timing code bug, motor driver overheating or incorrect wiring.

Test 6 : IR sensor failure (fault handling)

- **Purpose:** verify system fault detection and safe behaviour when an IR sensor disconnects.
- **How:** disconnect one IR sensor and observe system response.
- **Pass criteria:** system halts motion and raises an alert/indicator so operators know there's a fault.
- **Why it matters:** safety and robustness system should fail safe rather than continue in an unpredictable state.
- **Common failures:** fault not detected, no alert logged, system continues moving dangerously.

Test 7 : Motor direction test

- **Purpose:** confirm motor drivers and control code move the robot in commanded directions.
- **How:** send a forward command and observe the robot's movement.
- **Pass criteria:** robot moves straight forward as commanded and responds predictably.
- **Why it matters:** basic mobility test to ensure wiring, motor drivers, and control signals are correct.
- **Common failures:** reversed polarity (motors run backward), uneven wheel speeds, insufficient torque.

Test 8 : ESP32 communication (Wi-Fi + API)

- **Purpose:** ensure the ESP32 can connect to Wi-Fi and send data to external services (Twilio API).
- **How:** connect ESP32 to network and trigger a data send to Twilio.
- **Pass criteria:** data reaches the Twilio API and Twilio confirms message delivery.
- **Why it matters:** confirms reliable cloud communication for notifications and remote logs.
- **Common failures:** Wi-Fi auth failure, API key/config errors, HTTP library issues.

Test 9 : System integration

- **Purpose:** verify all modules (sensors, actuators, comms, control logic) work together as a complete system.
- **How:** connect all modules, run a typical use-case scenario and observe end-to-end behavior (detection → action → notification → logging).
- **Pass criteria:** synchronized operation with stable integration (no module conflicts, correct sequencing).
- **Why it matters:** unit components may work individually but integration ensures the whole workflow functions reliably.
- **Common failures:** timing/race conditions, data format mismatches, resource contention (serial/UART), or unhandled error propagation.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSIONS

The EcoSentinel project marks an important advancement in integrating robotics, IoT, and automation for sustainable waste management. The system combines line-following navigation, RFID/NFC-based waste detection, and automated SMS notifications through the Twilio API to create a smart, autonomous waste-monitoring solution. Its hardware components including IR sensors, Arduino Nano, L298N motor driver, NEO-6M GPS module, conveyor mechanism, and ESP32 for wireless communication work together to achieve accurate navigation, reliable tag detection, and seamless cloud connectivity. A key accomplishment of EcoSentinel is its ability to automatically identify violators through NFC-tagged waste and instantly notify them via SMS, promoting accountability and responsible behavior. During testing, the robot demonstrated stable operation, precise line tracking, dependable communication, and effective waste collection simulation, making it suitable for use in public spaces like parks, campuses, and institutional areas. The design's affordability, use of open-source platforms, and modular structure make it scalable for larger deployments. Overall, EcoSentinel successfully meets its objectives of automated waste identification, collection, and violator alert generation while contributing to eco-awareness, smart city development, and sustainable environmental practices.

8.2 FUTURE SCOPE

While EcoSentinel has achieved its fundamental goals, there remains substantial scope for improvement and expansion. The following future enhancements are proposed to increase the efficiency, scalability, and intelligence of the system:

- **Artificial Intelligence-Based Object Classification:** Currently, the robot relies on NFC stickers for waste identification. Future versions could integrate camera modules and machine learning algorithms to automatically detect and classify waste (plastic, metal, organic, etc.) without requiring NFC tags. This would enhance automation and eliminate the need for manual tagging.
- **Solar-Powered Operation:**The system presently operates using rechargeable batteries. Incorporating solar panels could make the robot more sustainable and environmentally friendly by reducing dependency on electrical charging, allowing for longer operational hours and reduced maintenance costs.

- **Cloud-Based Data Logging:**cloud storage services such as Firebase or AWS IoT Core would allow the robot to upload data (e.g., GPS location) a centralized dashboard. Authorities could use this data for analytics, violation tracking, and smart waste management planning.
- **Integration with Municipal Waste Systems:**Future enhancements could enable the system to automatically communicate with municipal databases. Each violation could be recorded against the responsible citizen's ID, enabling automated fine generation or warning systems through government portals.
- **Enhanced Navigation and Path Planning:**By using ultrasonic sensors, LIDAR, or computer vision, the robot could navigate more dynamically, avoiding obstacles and recalculating routes. This would allow it to operate in complex environments with unpredictable obstacles or varying terrains.

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APPENDIX A

Listing 1: Motor and Sensor Pin Configuration

```
    const int ENA = 11, IN1 = 10, IN2 = 9;
    const int ENB = 6,  IN3 = 8,  IN4 = 7;

    #define s 90      // Base speed
    #define t 255     // Turning speed

    const int IRSensorLeft = 3;
    const int IRSensorRight = 2;

    const int trig1 = 12, echo1 = 13;
    const int trig2 = 4,  echo2 = 5;

    const int switchPin = A0;
    const int buzzer = A1;
```

Listing 2: Ultrasonic Distance Measurement Function

```
long getDistance(int trig, int echo) {
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);

  long duration = pulseIn(echo, HIGH, 20000);
  if (duration == 0) return 0;

  return (duration * 0.034) / 2;
}
```

Listing 3: Obstacle Detection Logic

```
long d1 = getDistance(trig1, echo1);
long d2 = getDistance(trig2, echo2);

if (d1 <= 25 || d2 <= 25) {
  stopMotors();
  tone(buzzer, 1000, 200);
```

```
  return;
}
```

Listing 4: Line Following Logic

```
bool L = digitalRead(IRSensorLeft);
bool R = digitalRead(IRSensorRight);

if (L == LOW && R == LOW)
    moveForward();
else if (L == LOW && R == HIGH)
    turnRight();
else if (L == HIGH && R == LOW)
    turnLeft();
else
    stopMotors();
```

Listing 5: Motor Control Functions

```
void moveForward() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  analogWrite(ENA, s);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
  analogWrite(ENB, s);
}

void turnRight() {
  digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
  analogWrite(ENA, t);
  digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
  analogWrite(ENB, t);
}

void turnLeft() {
  digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
  analogWrite(ENA, t);
  digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
  analogWrite(ENB, t);
}

void stopMotors() {
```

```
  analogWrite(ENA, 0);
  analogWrite(ENB, 0);
}
```

Listing 6: Blynk Configuration

```
#define BLYNK_AUTH_TOKEN "23apIo5Mjijir1KQu_iFTnySqh59ljkT"
#define BLYNK_TEMPLATE_ID "TMPL3AEMFw0IT"
#define BLYNK_TEMPLATE_NAME "SMART WASTE COLLECTOR"
```

Listing 7: Blynk Configuration

```
#define BLYNK_TEMPLATE_ID "TMPL3AEMFw0IT"
#define BLYNK_TEMPLATE_NAME "SMART WASTE COLLECTOR"

#include <Wire.h>
#include <Adafruit_PN532.h>
#include <TinyGPSPlus.h>
#include <HardwareSerial.h>
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
```

Listing 8: Pin Definitions

```
#define SDA_PIN 21
#define SCL_PIN 22

#define GPS_RX 16
#define GPS_TX 17

#define SERVO_PIN 18
#define LASER_PIN 19
#define BUZZER 2
#define RED_LED 25
#define GREEN_LED 26

#define IN1 32
#define IN2 33
#define IN3 27
#define IN4 14
```

Listing 9: Object Initialization

```
Adafruit_PN532 nfc(SDA_PIN, SCL_PIN);
HardwareSerial gpsSerial(2);
TinyGPSPlus gps;

char auth[] = "23apIo5Mjijir1KQu_iFTnySqh59ljkT";
char ssid[] = "Meghana";
char pass[] = "meghana123";

BlynkTimer timer;
```

Listing 10: System Flags and Function Prototypes

```
bool gpsConnected = false;
bool gpsAttempted = false;
unsigned long lastGPSTryTime = 0;

void checkAllSensors();
bool checkGPS(int attempts);
bool checkRFID();
void runMotors(bool run);
void handleError();
void sendToBlynk(String tagID, float lat, float lon);
void beep(int count, int delayMs);
void servoRotate();
void tryReconnectGPS();
void blinkGreenIfNoGPS();
```

Listing 11: Setup Function

```
void setup() {
  Serial.begin(115200);
  Wire.begin(SDA_PIN, SCL_PIN);
  gpsSerial.begin(9600, SERIAL_8N1, GPS_RX, GPS_TX);

  pinMode(LASER_PIN, INPUT);
  pinMode(BUZZER, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(GREEN_LED, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
```

```
  pinMode(IN4, OUTPUT);

  ledcSetup(5, 50, 16);
  ledcAttachPin(SERVO_PIN, 5);

  Blynk.begin(auth, ssid, pass);

  nfc.begin();
  if (!nfc.getFirmwareVersion()) {
    handleError();
  }
  nfc.SAMConfig();

  checkAllSensors();

  timer.setInterval(500L, blinkGreenIfNoGPS);
  timer.setInterval(15000L, tryReconnectGPS);
}
```

Listing 12: Main Loop

```
void loop() {
  Blynk.run();
  timer.run();

  if (digitalRead(LASER_PIN) == LOW) {
    uint8_t uid[7], uidLength;
    if (nfc.readPassiveTargetID(PN532_MIFARE_ISO14443A, uid,
       &uidLength)) {
      String tagID = "";
      for (int i = 0; i < uidLength; i++)
        tagID += String(uid[i], HEX);
      tagID.toUpperCase();

      while (gpsSerial.available())
        gps.encode(gpsSerial.read());

      float lat = gps.location.lat();
      float lon = gps.location.lng();

```

```
      sendToBlynk(tagID, lat, lon);
      servoRotate();
      delay(1000);
    }
  }
}
```

Listing 13: Sensor Checking

```
void checkAllSensors() {
  bool gpsOk = checkGPS(3);
  bool rfidOk = checkRFID();

  if (rfidOk && gpsOk) {
    digitalWrite(GREEN_LED, HIGH);
    runMotors(true);
  } else if (rfidOk) {
    runMotors(true);
  } else {
    handleError();
  }
}
```

Listing 14: GPS Connectivity Check

```
bool checkGPS(int attempts) {
  for (int i = 0; i < attempts; i++) {
    unsigned long start = millis();
    while (millis() - start < 10000) {
      if (gpsSerial.available()) {
        gps.encode(gpsSerial.read());
        if (gps.location.isValid()) {
          gpsConnected = true;
          return true;
        }
      }
    }
  gpsConnected = false;
  return false;
}
```

Listing 15: Motor Control Functions

```
void runMotors(bool run) {
  if (run) {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
  } else {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
  }
}
```

Listing 16: Sending Data to Blynk

```
void sendToBlynk(String tagID, float lat, float lon) {
  Blynk.virtualWrite(V0, tagID);
  Blynk.virtualWrite(V1, lat);
  Blynk.virtualWrite(V2, lon);
}
```

Listing 17: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>EcoSentinal - Pay Penalty</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="card">
    <h1>       Penalty Notice</h1>
    <p>You have littered waste at <b>Jayanagar Bangalore Park
       </b>.</p>
    <p>Please pay your challan to avoid further action.</p>

    <div class="qr-section">
      <img src="qr.png" alt="QR Code to Pay" class="qr">
```

```
      <p>Scan QR to pay</p>
    </div>

    <button id="payBtn">Pay Now</button>
    <br><br>

    <!-- Button for admin redirection -->
    <button onclick="window.location.href='admin.html'">Go to
        Admin Page</button>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

Listing 18: script.js

```
const challan = {
  userId: "User123",
  location: "Jayanagar Bangalore Park",
  status: localStorage.getItem("penaltyStatus") || "Unpaid"
};

// Handle Pay Now button
const payBtn = document.getElementById("payBtn");
if (payBtn) {
  payBtn.addEventListener("click", () => {
    challan.status = "Paid";
    localStorage.setItem("penaltyStatus", "Paid");
    alert("    Payment Successful! Thank you.");
    window.location.reload();
  });
}

// Fill Admin Table (admin.html)
const userTable = document.getElementById("userTable");
if (userTable) {
  const row = document.createElement("tr");
  row.innerHTML = `
    <td>${challan.userId}</td>
```

```
    <td>${challan.location}</td>
    <td style="color:${challan.status === "Paid" ? "green" :
       "red"};">
      ${challan.status}
    </td>
  `;
  userTable.appendChild(row);
}
```

Listing 19: style.css

```
body {
  font-family: Arial, sans-serif;
  background: linear-gradient(135deg, #56ccf2, #2f80ed);
  margin: 0;
  padding: 0;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}

.card {
  background: white;
  border-radius: 15px;
  padding: 25px;
  box-shadow: 0px 8px 20px rgba(0,0,0,0.2);
  text-align: center;
  width: 400px;
  max-width: 90%;
}

h1 {
  color: #2f80ed;
}

.qr-section {
  margin: 20px 0;
}

```

```
.qr {
  width: 150px;
  border: 2px solid #eee;
  border-radius: 10px;
}

button {
  background: #2f80ed;
  color: white;
  border: none;
  padding: 12px 25px;
  border-radius: 8px;
  cursor: pointer;
  font-size: 16px;
}

button:hover {
  background: #1c5ed6;
}

table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 15px;
}

th, td {
  border: 1px solid #ddd;
  padding: 10px;
}

th {
  background: #2f80ed;
  color: white;
}

td {
  text-align: center;
}
```

Listing 20: admin.html

```
    <!DOCTYPE html>
    <html lang="en">
    <head>
    <meta charset="UTF-8">
     <title>EcoSentinal - Admin Dashboard</title>
     <link rel="stylesheet" href="style.css">
    </head>
    <body>
    <div class="card">
    <h1>          Admin Dashboard</h1>
    <p>Monitor challan payments.</p>

    <table>
      <thead>
        <tr>
          <th>User ID</th>
          <th>Location</th>
          <th>Status</th>
        </tr>
      </thead>
      <tbody id="userTable">
        <!-- Filled by JS -->
      </tbody>
    </table>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

Listing 21: Blynk and Twilio Configuration

```
BLYNK_AUTH_TOKEN = "23apIo5Mjijir1KQu_iFTnySqh59ljkT"
PINS = ["V0", "V1", "V2"]

CHALLAN_BASE = "https://kkhyathikiran.github.io/
   challanEcosentinel/"

TWILIO_ACCOUNT_SID = "00000000000000000000000000000000"
```

```
TWILIO_AUTH_TOKEN = "00000000000000000000000000000000"
TWILIO_FROM_NUMBER = "000000000000000"
TWILIO_TO_NUMBER = "+0000000000"
```

Listing 22: Fetching Latest Blynk Data

```
def fetch_all_latest(auth_token, pins):
    """Fetch each pin individually from Blynk Cloud."""
    data = {}
    for pin in pins:
        url = f"https://blynk.cloud/external/api/get?token={auth_token}&{pin}"
        try:
            resp = requests.get(url, timeout=5)
            print(f"DEBUG [{pin}]     {resp.status_code} | {resp.text}")
            if resp.status_code == 200 and resp.text not in ("", "null", "None"):
                data[pin] = resp.text.strip()
            else:
                data[pin] = ""
        except Exception as e:
            print(f"    Error fetching {pin}: {e}")
            data[pin] = ""
    return data
```

Listing 23: Challan Link Generator

```
def make_challan_link(base_url, tag_id, lat, lon, violation="throwing_dust"):
    """Generate clickable challan URL with parameters."""
    ts = datetime.utcnow().isoformat(timespec='seconds') + "Z"

    params = {
        "tag": tag_id,
        "lat": lat or "",
        "lon": lon or "",
        "violation": violation,
        "ts": ts
    }
```

```
    return base_url.rstrip("/") + "/?" + urlencode(params)
```

Listing 24: Sending SMS Using Twilio

```
def send_sms(challan_link, tag_id, lat, lon):
    """Send challan link to user via SMS."""
    try:
        client = Client(TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN)
        body = (
            f"Violation Detected!\n"
            f"Tag ID: {tag_id}\n"
            f"Location: https://maps.google.com/?q={lat},{lon}\n"
            f"Challan Link: {challan_link}"
        )

        message = client.messages.create(
            body=body,
            from_=TWILIO_FROM_NUMBER,
            to=TWILIO_TO_NUMBER
        )

        print(f"    SMS sent successfully! SID: {message.sid}")
    except Exception as e:
        print(f"    Failed to send SMS: {e}")
```

Listing 25: Main Monitoring Loop

```
if __name__ == "__main__":
    last_tag_id = None
    POLL_INTERVAL = 5
    print("Monitoring Blynk")

    try:
        while True:
            data = fetch_all_latest(BLYNK_AUTH_TOKEN, PINS)

            if data:
                tag_id = data.get("V0") or data.get("v0")
```

```
                lat = data.get("V1") or data.get("v1")
                lon = data.get("V2") or data.get("v2")

                tag_id = tag_id.strip() if isinstance(tag_id,
                    str) else tag_id
                lat = lat.strip() if isinstance(lat, str)
                   else lat
                lon = lon.strip() if isinstance(lon, str)
                   else lon

                if tag_id and tag_id not in ("null", "None",
                   "") and tag_id != last_tag_id:
                    print(f"\nNew Tag Detected: {tag_id}")
                    print(f"Latitude: {lat}, Longitude: {lon}
                       ")

                    challan_link = make_challan_link(
                       CHALLAN_BASE, tag_id, lat, lon)
                    print(f"Generated Challan Link: {
                       challan_link}")

                    send_sms(challan_link, tag_id, lat, lon)
                    last_tag_id = tag_id
                else:
                    print(".", end="", flush=True)

            else:
                print("No data received from Blynk.")

            time.sleep(POLL_INTERVAL)

    except KeyboardInterrupt:
        print("\nStopped by user.")
```